Chaos Game Representation(CGR)

A PROJECT REPORT

for the course

21BIO201- Introduction to Robotics

TEAM – E4

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ABSTRACT

The coronavirus, also known as Covid-19 virus, has been confirmed in approximately 7 million people in 213 countries, with nearly 400,000 fatalities. For planning, containment, and treatment of such significant epidemics, the categorization and origin of the viral genomic sequence are required. We provide two alignment-free approaches that combine with CGR to conduct clustering analysis and generate a phylogenetic tree as a result of the aforementioned necessity. We assign a matrix to each DNA sequence and then define the distance between DNA sequences as the distance between their corresponding matrices. The phylogenetic analysis of coronavirus sequences is carried out using these techniques. With our method, you can easily analyse and annotate genomes and their phylogenetic connections. We demonstrate the ability of our alignment-free approaches to identify coronaviruses' closest genetic cousins.

INTRODUCTION

The Chaos Game Representation (CGR) is a method for visualizing the structure of DNA sequences. The CGR method is based on the concept of iteratively applying a set of simple rules, or "chaos game," to the sequence of nucleotides in a DNA fragment. The end result is a fractal pattern that can be used to identify patterns and regularities in the DNA sequence. This pattern is created by plotting the locations of the nucleotides at each iteration of the chaos game, where the location is determined by the particular set of rules used in the game. The CGR method has been used to identify patterns in a wide range of DNA sequences, including those from different organisms and different regions of the genome. It is particularly useful for identifying patterns in non-coding regions of DNA, such as in introns, promoters and enhancers.

Leigh syndrome, also known as subacute necrotizing encephalomyelopathy, is a rare, inherited disorder that affects the central nervous system, specifically the brain and spinal cord. The disorder typically presents in infants and young children, and it leads to progressive neurological deterioration and, eventually, death. The symptoms of Leigh syndrome can include developmental delays, muscle weakness, loss of motor skills, and difficulty with breathing and feeding. The condition is caused by mutations in genes that are involved in the metabolism of certain molecules, such as glucose and lipids. Some of the most common genetic causes of Leigh syndrome are mutations in the mitochondrial DNA, which are inherited from the mother.

Leigh syndrome is a rare disorder and the prevalence is not known. However, it is estimated to affect 1 in every 40,000-50,000 individuals. The symptoms and progression of the disorder can vary widely, even among individuals with the same genetic mutation, which makes it difficult to predict the course of the disease for an individual patient. There is no cure for Leigh Syndrome, The management mainly focused on treating symptoms and preventing complications, as well as providing supportive care for patients and families.

In this project data has been gathered on two categories of data - hCov-19 and BetaCov-19 sequences and also for human and various animal genome sequences (Gallus, Goat, Gorilla, Human, Lemur, Mouse, Opossum, Rabbit, Rat)

LITERATURE SURVEY

[1] In this paper, results show that method used can accurately classify different genomic sequences. In terms of classification accuracy, method used is basically the same as the state-of-the art Clustal X and compare with the traditional Clustal X phylogenetic tree construction method , our method is much faster. Furthermore, dendrogram construct method can be widely applicable for various kinds of organisms. This research may contribute to reveal the biological evolution process to some extent, as well as promote the further development of bioinformatics. We may make efforts in our future work to provide a webserver for the methods presented in this paper. All the codes in this paper are written in python and can be available upon request.

[2] The present article describes the algorithms for generation of CGRs of the primary sequences of the members of different protein families and quantitative measurement of the bias, if any, in such CGR patterns. It has been demonstrated that different protein families exhibit distinct patterns in their CGRs with characteristic grid counts. It is, therefore, possible to use the grid counts as diagnostic features of such protein families for identification of new members of the families. In 12-vertex CGRs of proteins, the patterns are dictated not only by the frequencies of occurrence of different residues but also by the statistical bias in the occurrence of di-, tri-, or higher order peptides in the amino acid sequences of proteins.

[3] This paper discuss about A new algorithm that uses information from chaos game representation of genome sequences for finding all local alignments between the sequences has been developed. Fast comparisons can be made between sequences of megabase size using a Pentium IV machine. As far as the speed of alignment is concerned, the program, in its present state does not offer any major improvements over MUMmer, but it is possible that the method can be implemented more efficiently through better programming inputs. Addition of the possibility of large scale sequence alignment to the existing repertoire of alignment-free sequence analysis possibilities from chaos game representation, positions CGR as a powerful quantitative sequence analysis tool.

DNA Sequence:

1. **For hCov-19 and BetaCov-19**

>Gallus(Red Junglefowl)

ATGGTGCACTGGACTGCTGAGGAGAAGCAGCTCATCACCGGCCTCTGGGGGAAGGT
CAATGTGGCCGAATGTGGGGCCGAAGCCCTGGCCAG



>Goat

ATGCTGACTGCTGAGGAGAAGGCTGCCGTCACCGGCTTCTGGGGCAAGGTGAAAGT
GGATGAAGTTGGTGCTGAGGCCCTGGGCAG



>Gorilla

ATGGTGCACCTGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGT
GAACGTGGATGAAGTTGGTGGTGAGGCCCTGGGCAGG



>Human
ATGGTGCACCTGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGT
GAACGTGGATTAAGTTGGTGGTGAGGCCCTGGGCAG

>Lemur
ATGACTTTGCTGAGTGCTGAGGAGAATGCTCATGTCACCTCTCTGTGGGGCAAGGT
GGATGTAGAGAAAGTTGGTGGCGAGGCCTTGGGCAG



>Mouse
TCTCTTGCCTGTGGGCAAAGGTGAACCCCGATGAAGTTGGTGGTGAGGCCCTGGGC
AGGATGGTGCATCTGTCCAGTGAGGAGAAGTCTGCGGT



>Opossum

ATGGTGCACTTGACTTCTGAGGAGAAGAACTGCATCACTACCATCTGGTCTAAGGT
GCAGGTTGACCAGACTGGTGGTGAGGCCCTTGGCAG



>Rabbit

CACTGCCCTGTGGGGCAAGGTGAATGTGGAAGAAGTTGGTGGTGAGGCCCTGGGCA
TGGTGCACCTAACTGATGCTGAGAAGGCTACTGT



>Rat

TAGTGGCCTGTGGGGAAAGGTGAACCCTGATAATGTTGGCGCTGAGGCCCTGGGCA
GATGGTGCACCTGACTCCTGAGGAGAAGTCTGCCGT



2. For Leigh Syndrome

Leigh Syndrome is a rare and progressive neurological disorder that primarily affects the central nervous system, including the brain and spinal cord. The disorder is characterized by the degeneration of specific areas of the brain, particularly the mitochondria. Symptoms typically appear in the first year of life and can include developmental delays, poor muscle tone, difficulty with movement, seizures, and vision and/or hearing loss. The disorder can lead to serious complications and can be fatal, usually between ages of 2 and 4.
It is inherited in an autosomal recessive pattern, which means that both copies of the gene in each cell have mutations. The mutations cause the body's cells to produce damaged or nonworking proteins or enzymes that are involved in the metabolism of the cell. This can lead to the death of cells, particularly cells in the brain and muscles.

Leigh Syndrome is a rare disorder and the exact incidence is unknown. However, it is estimated that it affects 1 in 40,000 to 60,000 live births.

The disorder is caused by mutations in several genes, the most common being the mitochondrial DNA-encoded NADH-ubiquinone oxidoreductase subunit gene (MT-NADH-QR), the gene encoding the mitochondrial DNA polymerase γ (POLG) and the gene encoding the mitochondrial transcription factor A (TFAM). These genes play critical roles in maintaining the function of the mitochondria.
Symptoms of Leigh Syndrome can vary depending on the specific gene mutation, but they typically include:

1. Developmental delays
2. Weak muscle tone (hypotonia)
3. Difficulty with movement and coordination (ataxia)
4. Seizures
5. Vision and/or hearing loss
6. Breathing difficulties and feeding problems
7. Inflammation of the brain (encephalitis)
8. Loss of developmental milestones
9. -Lactic acidosis
10. Diagnosis is based on the combination of clinical features, imaging, and metabolic tests. Genetic testing can also be done to confirm the diagnosis.

There is no cure for Leigh Syndrome, and treatment is mainly supportive and symptomatic. It aims to manage symptoms and prevent complications. Treatment may include physical therapy, speech therapy, and occupational therapy to help with developmental delays and muscle weakness, anticonvulsants to control seizures, and treatment to manage lactic acidosis. A multidisciplinary team of specialists including neurologists, geneticists, metabolic specialists, and other healthcare professionals can work together to provide comprehensive care.

For Leigh syndrome the genome is taken from NCBI website (fasta file)

NC_012920.fasta
https://www.ncbi.nlm.nih.gov/nuccore/NC_012920.1?report=fasta

CODE-1:

```
import collections
from matplotlib import pyplot as plt
from matplotlib import cm
import math

f = open("NC_012920.fasta")
s1 = f.read()
data = "".join(s1.split("\n")[1:])

def count_kmers(sequence, k):
    d = collections.defaultdict(int)
    for i in range(len(data)-(k-1)):
        kmer = sequence[i:i+k]
        if "N" in kmer:
            continue
        d[kmer] +=1
    return d

def probabilities(kmer_count, k):
    probabilities = collections.defaultdict(float)
    N = len(data)
    for key, value in kmer_count.items():
        probabilities[key] = float(value) / (N - k + 1)
    return probabilities

def chaos_game_representation(probabilities, k):
    array_size = int(math.sqrt(4**k))
    chaos = [[0 for x in range(array_size)] for y in range(array_size)]
    for key, value in probabilities.items():
        maxx = array_size
        maxy = array_size
        posx = 1
        posy = 1
        for char in key:
            if char == "T":
                posx += maxx // 2
            elif char == "C":
                posy += maxy // 2
            elif char == "G":
                posx += maxx // 2
                posy += maxy // 2
            maxx = maxx // 2
            maxy = maxy // 2
        chaos[posy-1][posx-1] = value
    return chaos

f3 = count_kmers(data, 3)
f4 = count_kmers(data, 4)
print(f3)
print(f4)

f3_prob = probabilities(f3, 3)
f4_prob = probabilities(f4, 4)

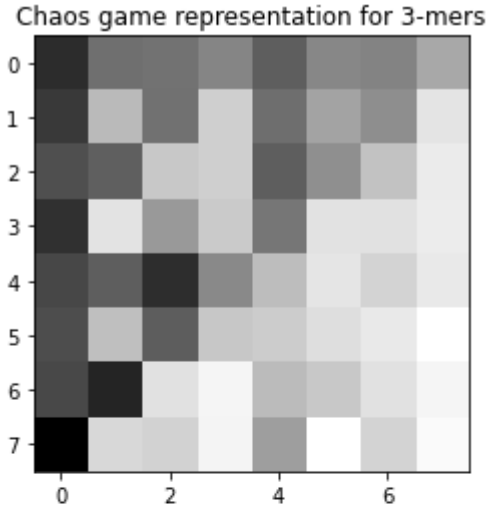
chaos_k3 = chaos_game_representation(f3_prob, 3)
plt.title('Chaos game representation for 3-mers')
plt.imshow(chaos_k3, interpolation='nearest', cmap=cm.gray_r)
plt.show()

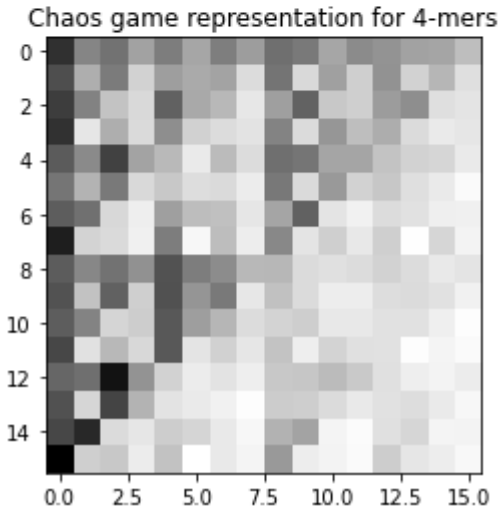
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plt.title('Chaos game representation for 4-mers')
plt.imshow(chaos_k4, interpolation='nearest', cmap=cm.gray_r)
plt.show()
```

OUTPUT-1:

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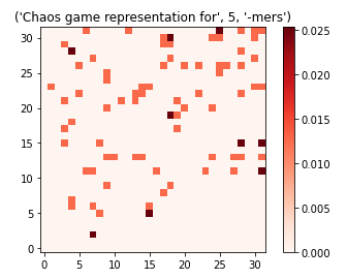


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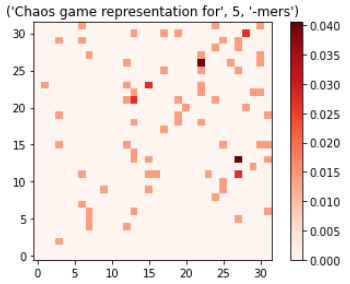
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OUTPUT-2

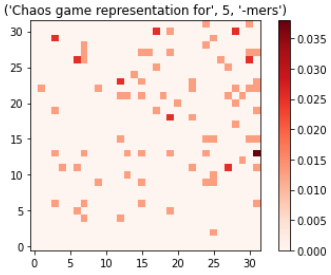
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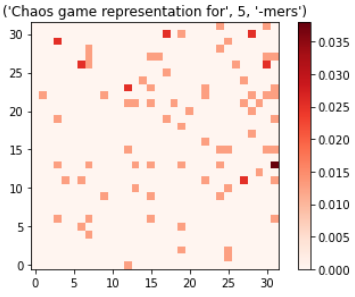
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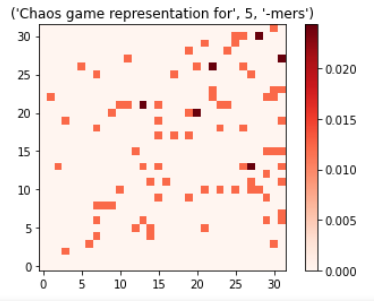
Gorilla.fasta



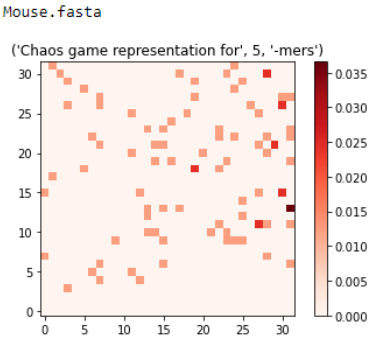
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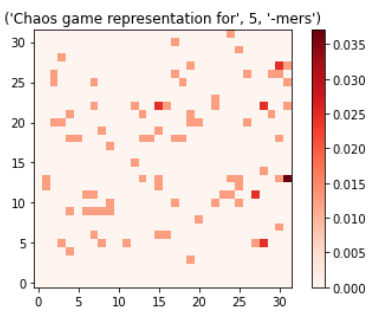
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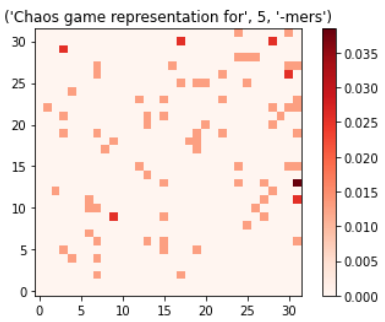
Mouse.fasta



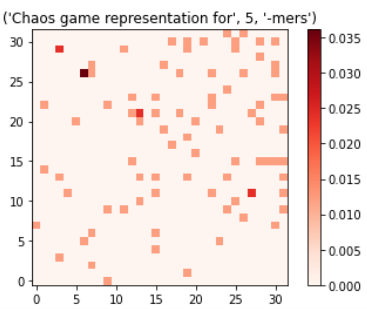
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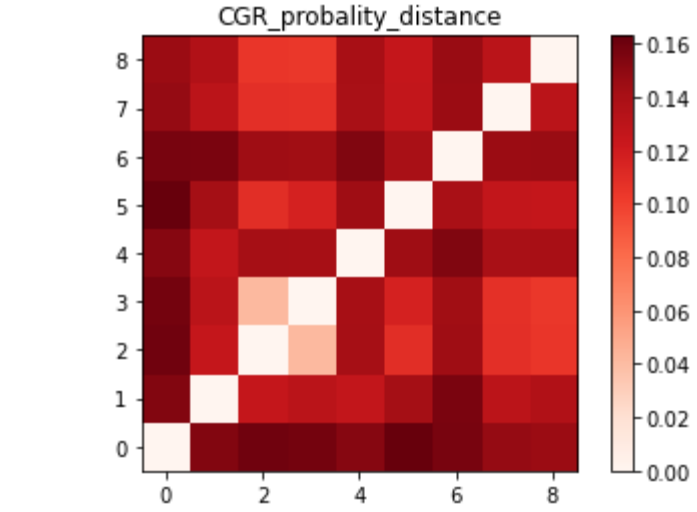


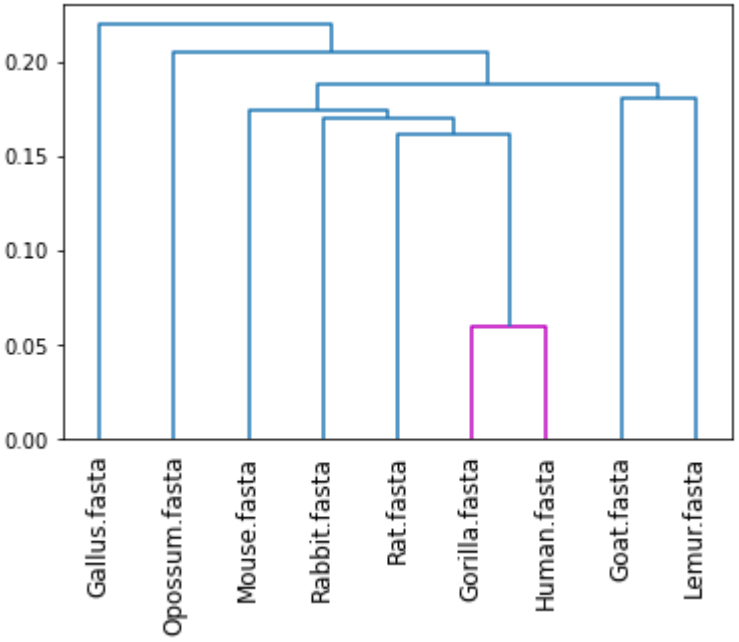
Rabbit.fasta



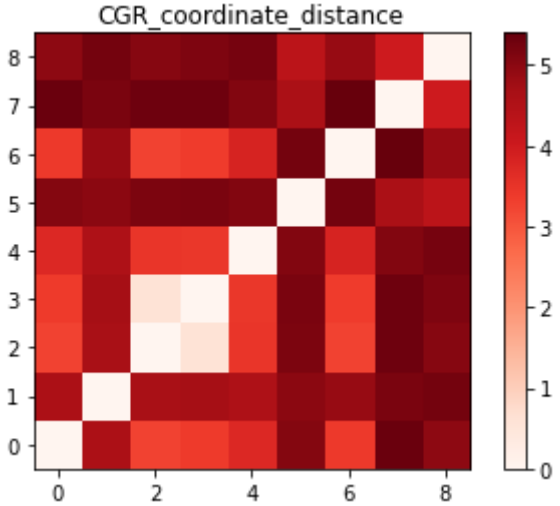
Rat.fasta







Final CGR Probability



CONCLUSION

- No treatments can alter the course of Leigh Syndrome

- There is no treatment to stop or reverse Leigh Syndrome, However there are some medications that can help keep symptoms under control.

- But we can identify Global and local patterns for Leigh Syndrome

- After the demonstration we found out that Humans and Gorillas are identified to be coronaviruses' closest genetic cousins.

- From polygenetic tree we found that not only Humans and Gorillas even Rat, Gorilla and Rat, Human are most closely related.

- From polygenetic tree we found that opossum-Rabbit , Gallus-Rabbit and Gorilla-Rabbit are least closely related

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[4] https://www.youtube.com/watch?v=oaG7ocRs5Ek